



NAKKA HEMANTH DATTA

Engineering Physics | EP23B028

CGPA: 8.45 | GitHub | LinkedIn

Education

- Indian Institute of Technology Madras** | Chennai, TN
B. Tech - Engineering Physics | Jul '23 – Jul '27
- JNV, Pondicherry** | Pondicherry, PY
Class XII (CBSE) | 95.80% | 2023

Professional Experience

Wells Fargo | Bangalore, India
Software Engineering Intern | Full Stack & AI Systems | May '26 – Jul '26

- Deployed a containerized **3-service trade-processing stack** (FastAPI, React, Redis) on Kubernetes with HPA autoscaling & liveness/readiness probes, raising throughput to **10,000+ trade confirmations/hour** & replacing manual review workflows.
- Architected the underlying **extraction engine** using **Strategy Design Patterns, PyMuPDF, and Redis caching**, automating **JSON-to-SQL schema normalization** & sub-second trade database reconciliation.
- Developed a **real-time React review workbench** featuring **SSE-based status streaming**, split-screen verification, embedded PDF inspection, & **pagination** for rapid exception handling.
- Developed a **trade management system** using **Angular 16, ASP.NET Core 8, Dapper, & SQL Server**, optimizing data access via batched SOR retrieval, in-memory caching, & transactional status synchronization.
- Implemented **API error handling, structured logging, & unit tests** across FastAPI services, covering validation failures, Redis/K8s outage paths, & increasing test coverage for the trade-confirmation flow.

CV. OMSAI INFORMASI TEKNOLOGI | Remote, Singapore
Software Engineer (Contract) | Feb '26 – May '26

- Partnered directly with an **Oman-based enterprise client** to build, iterate, & deploy a production AI document intelligence system on a **cloud VM**, driving end-to-end solution delivery.
- Engineered **RESTful API integrations** & a **multimodal Corrective RAG (CRAG) pipeline** using **LangGraph, OpenAI embeddings & LLMs, ChromaDB (vector + BM25), semantic chunking**, & a **reranking engine** with query-rewrite/re-retrieval on low-confidence hits, achieving **100% table query accuracy across a 250+ enterprise document benchmark**.
- Built a **React frontend** with **live citation highlighting & lazy-loaded document rendering**, using **Ragas framework context metrics** to trim prompt token usage by **35%**, lowering **API latency** & reducing per-request cost.

Technical Skills

- Languages:** Python, C/C++, SQL, Matlab, Mathematica
- Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, Ragas, FAISS, ChromaDB, Unsloth, FastAPI, React, Angular 16, ASP.NET Core, Dapper, SQL Server, Docker, Kubernetes, Redis, Celery, Streamlit, pandas, Git
- Core Skills:** DSA, Operating Systems, Distributed Systems, Concurrency & Multithreading, SQL Indexing & Query Optimization, Machine Learning, Deep Learning, NLP, Computer Vision, RAG, LLM Fine-Tuning (PEFT/LoRA), LLMops & Observability

Relevant Coursework

- | | |
|---------------------------------|-------------------------------|
| • Data Structures & Algorithms | • Computer Organisation |
| • Database Management Systems | • Modern App Development |
| • Probability & Statistics | • Natural Language Processing |
| • Intro to Machine Learning | • Computer Vision |
| • Introduction to Deep Learning | • Computational Physics |

Position of Responsibility

CFI: Sahaay Social Innovation Club
Project Member (Gesturize) | Apr '24 – Mar '25

- Co-invented an Indian Sign Language (ISL) translation system on low edge devices using **Siamese Networks, MediaPipe, & interpolation** (Patent App. No: 202541022908), leading 60+ **NSS IIT Madras** volunteers for open-source dataset contribution.

Mentorship

Avanti Fellows & EE Intern Buddy Program
Student Mentor | Sep '23 – Aug '26

- Mentored underprivileged JEE aspirants by giving back as an Avanti Fellows alumnus & guided junior EE students as Mentor.

Projects

Distributed High-Performance N-Body Simulation Engine | Jan '26 – May '26
Under Prof. Yasir Iqbal | PH5730

- Engineered a **gravitational N-body simulator** in **C++20/Python** as a numerical-optimization ladder: naive $\mathcal{O}(N^2)$ pairwise forces $\rightarrow$ vectorized NumPy kernel $\rightarrow$ **Barnes-Hut tree** ($\mathcal{O}(N \log N)$) with opening-angle θ approximation (rel. error $< 10^{-12}$).
- Vectorized the force kernel for a $\sim 78\times$ **speedup** over the naive loop ($N = 256$) & **parallelized Barnes-Hut tree traversal** across **8 multiprocessing workers** for a $\sim 3.0\times$ further speedup ($N = 8192$), tuning θ against accuracy.
- Built a **socket-based coordinator/worker protocol** with **raw-byte synchronization** ($< 0.1\%$ overhead) & **fail-stop checkpoints**, benchmarking 6 scaling experiments across N , worker count, & θ on compute/communication/accuracy tradeoffs.

Distributed Key-Value Store

C++ | Distributed Systems | CMake | Jul '26 – Aug '26

- Designed & implemented a **Redis-compatible distributed key-value store** in C++ with modular architecture comprising **Reactor-pattern networking layer, concurrent storage engine, durability subsystem, & Raft-based consensus module**.
- Built a **Reactor-pattern TCP server** with **worker thread pool**, implementing thread-safe in-memory storage, **WAL durability with validation**, snapshot-based recovery, serializable transactions, & **dual-layer TTL expiration** using a scheduler.
- Implemented **Raft consensus** with randomized leader election, heartbeat RPCs, **quorum-based replication**, & follower catch-up; validated correctness using a **15-test CTest suite** & a **benchmark client** measuring throughput & latency under concurrent workloads.

Colab Sync Control – Open-Source Developer Tool

VS Code Extension | Node.js | WebSockets | Colab API | Jul '26 – Present

- Built a VS Code extension enabling **bi-directional synchronization between local workspaces and remote Google Colab runtimes**, with automatic background syncing & conflict monitoring.
- Architected a **local daemon-based control plane** for workspace linking, remote session lifecycle management, & interactive shell access, integrating cloud compute into the native VS Code workflow.
- Developed a real-time dashboard for **CPU/RAM/disk/GPU telemetry & compute-unit tracking**, supporting T4/L4/A100 GPU & TPU runtimes.
- Published to the **VS Code Marketplace**, featuring **auto-reconnects, hash-based diffing**, & encrypted WebSocket channel.

Foundational Model Adaptation Pipeline

Under Prof. Sukhendu Das | CS6350 | Jul '25 – Dec '25

- Built a modular **model adaptation pipeline** for CLIP & DINOv2 supporting zero-shot, few-shot, & parameter-efficient fine-tuning workflows.
- Implemented **LoRA, BitFit, & Prefix Tuning** & optimized training to update only **0.7% of model parameters**, achieving **94.67% accuracy** with DINOv2.
- Designed a **two-stage training architecture** separating feature extraction & classifier optimization, enabling efficient experimentation across model & adaptation configurations.

Hackathons

2nd Prize, Mando AI Tool Dev Hackathon 2025

LangGraph | Claude LLM | FAISS | Azure | May '25 – Jun '25

- Engineered a full-stack document QA system supporting PDF, DOCX, & CSV, using LangGraph, VoyageAI embeddings, Claude LLM, & FAISS hybrid search across **2,000+ page documents**.

Round 3 Finalist, Appian Algnite Hackathon 2025

CLIP | Gemini API | FAISS | Azure | Mar '25 – Apr '25

- Built a multimodal product discovery platform using CLIP+FAISS hybrid search across **14K+ products**, with a Gemini-powered AI browser extension for automated e-commerce checkout.

Scholastic Achievements

- CBSE Certificate of Merit for top 0.1% rank (100/100) in CS, Class XII Boards 2023 & selected for Reliance UG Scholarship
- AIR 5476 in JEE Advanced 2023 (out of 2.5L+ candidates) & Top 1.42% in JEE Mains 2023 out of 12L+ candidates

Extra-Curriculars

- NSO Volleyball & state/national chess player & EEA frisbee champions